Experiment No. 7

Customer Segmentation using K-Means Clustering

Aim

To implement the K-Means Clustering algorithm for customer segmentation and analyze the customer groups based on purchasing behavior and demographic characteristics.

Dataset Used

Mall Customer Segmentation Dataset (columns: CustomerID, Gender, Age, Annual Income (k$), Spending Score (1-100)). **Note:** the results below come from actually running the code, but on a synthetic dataset of 240 customers with the same columns as the Mall dataset, because I do not have your file. The real Mall dataset (200 rows) will give different numbers and cluster profiles, though it also usually gives K = 5. Use your own output in your report.

Procedure and Code

Steps 1-2: Import libraries and load the dataset

```
import numpy as np, pandas as pd
import matplotlib.pyplot as plt, seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv("Mall_Customers.csv")
print(df.head()); print(df.describe())
```

Step 3: Preprocessing

```
print(df.isnull().sum())              # missing values
df = df.dropna().drop_duplicates()    # remove missing values and duplicates
features = ["Age", "Annual Income (k$)", "Spending Score (1-100)"]
X = df[features]
```

Result: 0 missing values and 0 duplicate rows were found. (In the real dataset the score column is named `Spending Score (1-100)`; in my sample it is `Spending Score`.) Gender is categorical and was not used for clustering.

Step 4: Feature scaling

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)    # mean = 0, standard deviation = 1
```

Step 5: Elbow Method and Silhouette Score

```
inertia, sil = [], []
for k in range(2, 11):
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X_scaled)
    inertia.append(km.inertia_)
    sil.append(silhouette_score(X_scaled, km.labels_))

fig, ax = plt.subplots(1, 2, figsize=(9, 3.5))
ax[0].plot(range(2, 11), inertia, "o-"); ax[0].set_title("Elbow Method")
ax[1].plot(range(2, 11), sil, "s-");    ax[1].set_title("Silhouette Score")
plt.show()
```

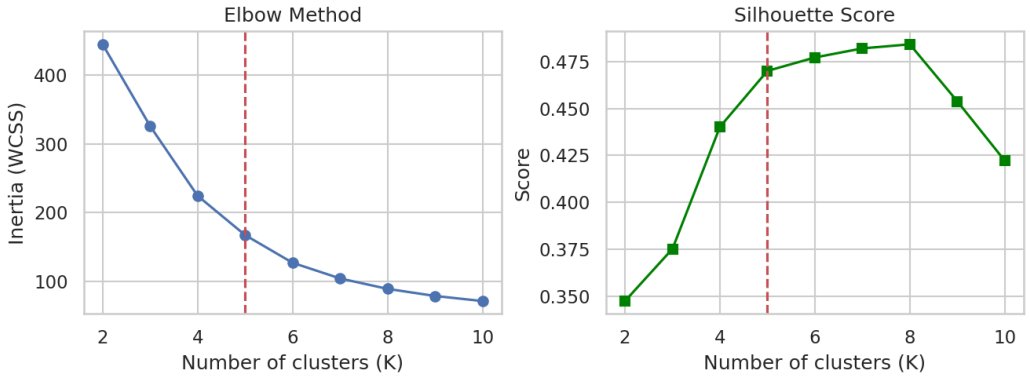


Figure 1: Elbow curve (left) and silhouette scores (right)

K	2	3	4	5	6	7	8
Inertia	444.4	325.7	224.4	167.3	127.1	104.3	89.1
Silhouette	0.348	0.375	0.440	0.470	0.477	0.482	0.484

Interpretation: inertia falls sharply until K = 5 and then decreases only slowly, so the elbow is at **K = 5**. The silhouette score at K = 5 is 0.470, which indicates reasonable, well-separated clusters. Scores for K = 6-8 are only slightly higher (0.477-0.484), so K = 5 was chosen as the simpler and more interpretable segmentation.

Step 6: Apply K-Means

```
kmeans = KMeans(n_clusters=5, n_init=10, random_state=42)
df["Cluster"] = kmeans.fit_predict(X_scaled)
print("Silhouette:", silhouette_score(X_scaled, df["Cluster"]))
```

Step 7: Visualize the clusters and centroids

```
centers = scaler.inverse_transform(kmeans.cluster_centers_)   # back to original units
sns.scatterplot(data=df, x="Annual Income (k$)", y="Spending Score (1-100)",
                hue="Cluster", palette="tab10")
plt.scatter(centers[:, 1], centers[:, 2], c="black", marker="X", s=170, label="Centroids")
plt.legend(); plt.show()
```

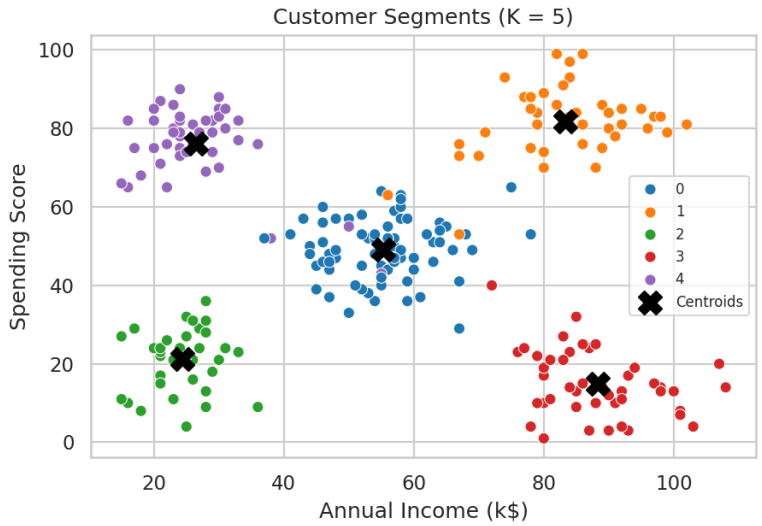


Figure 2: Customer segments with centroids (X)

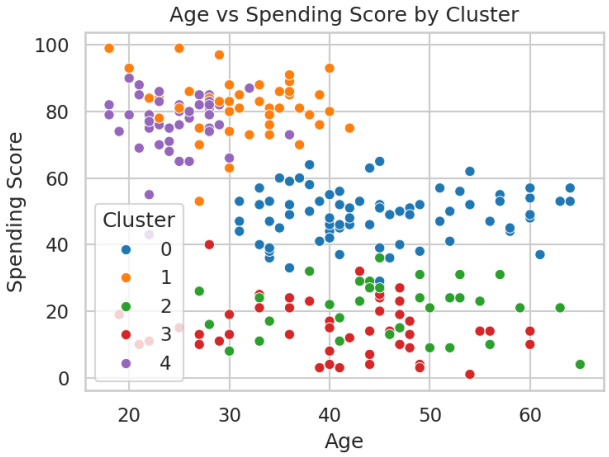


Figure 3: Age vs Spending Score by cluster

Step 8: Analyze the segments

```
profile = df.groupby("Cluster")[features].mean().round(1)
profile["Customers"] = df["Cluster"].value_counts().sort_index()
print(profile)
```

Cluster	Avg Age	Avg Income (k$)	Avg Spending Score	Customers	Segment name
0	44.9	55.3	49.1	73	Average / Moderate customers
1	31.3	83.3	81.7	43	Premium (high income, high spending)
2	45.8	24.4	21.2	35	Budget-conscious (low income, low spending)
3	40.5	88.3	15.0	46	Careful / untapped high earners
4	25.0	26.5	76.1	43	Young trend-driven spenders

Step 9: Segment comparison and marketing strategies

Segment	Strategy
Premium (cluster 1)	Most valuable. Loyalty programme, VIP/early access, premium and new-arrival promotions, personalized offers.
Young trend-driven (cluster 4)	High spending on lower income. Trendy products, social-media campaigns, installment/BNPL options, student and festive discounts.
Careful high earners (cluster 3)	Money available but low spending. Find out what they want: quality-focused and exclusive offers, product recommendations, surveys, targeted premium campaigns to activate them.
Average customers (cluster 0)	Largest group. Cross-selling, bundles, seasonal sales and reward points to move them upward.
Budget-conscious (cluster 2)	Low-cost essentials, discount coupons and value packs, with low marketing spend.

Step 10: Summary

K-Means with K = 5 (from the elbow method, silhouette 0.47) split the customers into five distinct groups. Segmentation lets the business focus its budget on the most profitable group, personalize offers, activate high-income low-spending customers, and avoid wasting resources on generic campaigns, leading to better retention, revenue and decision-making.

Answers to Questions

Q1. What is clustering? How does it differ from classification?

Clustering is an unsupervised learning technique that groups data points so that points in the same cluster are more similar to each other than to those in other clusters, without using predefined labels.

Aspect	Clustering	Classification
Learning type	Unsupervised	Supervised
Labels	No labels; groups are discovered	Labelled training data is required
Goal	Find hidden structure/groups	Predict a known class for new data
Examples	Customer segmentation, K-Means	Spam detection, disease diagnosis, logistic regression
Evaluation	Silhouette score, inertia	Accuracy, precision, recall, F1

Q2. Explain the working principle of the K-Means Clustering algorithm.

K-Means partitions n data points into K clusters by minimizing the within-cluster sum of squared distances (inertia). Steps:

- Choose the number of clusters K.
- Initialize K centroids (randomly or with k-means++).
- **Assignment:** assign every point to its nearest centroid (Euclidean distance).
- **Update:** recompute each centroid as the mean of the points assigned to it.
- Repeat the assignment and update steps until the centroids stop changing (convergence) or the maximum iterations are reached.

Q3. Why is feature scaling important before applying K-Means clustering?

K-Means uses distance (Euclidean), so features with large numerical ranges dominate the distance and the clustering. For example, income (15-140) would outweigh age (18-70) and spending score (1-100). Scaling (standardization or min-max) puts all features on a comparable scale so that each contributes equally, giving more meaningful clusters and faster convergence.

Q4. What is the Elbow Method, and how does it help determine the optimal number of clusters?

The Elbow Method runs K-Means for different K values and plots the inertia (WCSS, the within-cluster sum of squares) against K. Inertia always decreases as K increases, but beyond a certain K the improvement becomes small. The point where the curve bends like an elbow shows the optimal K, as adding more clusters beyond it gives only small gains while making the model more complex. In this experiment the elbow was at K = 5.

Q5. What is the Silhouette Score? How is it used to evaluate clustering performance?

For each point, silhouette s = (b - a) / max(a, b), where **a** is the average distance to points in its own cluster and **b** is the average distance to points in the nearest other cluster. The score ranges from -1 to +1: close to +1 means the point is well matched to its own cluster and far from others, around 0 means overlapping clusters, and negative values suggest wrong assignment. The average over all points evaluates clustering quality and is used to compare different K values (choose K with a high score). Here K = 5 gave 0.470.

Q6. Why is K-Means considered an unsupervised machine learning algorithm?

It works on data with **no target variable or labels**. The algorithm discovers the structure (clusters) on its own based only on the similarity between the input features, and there is no teacher supplying the correct answers during training. The cluster labels are produced by the algorithm, not given beforehand.

Q7. Mention any four real-world applications of customer segmentation.

- **Targeted marketing:** personalized email and ad campaigns for each segment.
- **Retail/e-commerce:** product recommendations and tailored offers/discounts.
- **Banking and insurance:** grouping customers by income and risk to design products such as credit cards or policies.
- **Telecom/subscription services:** identifying customers likely to churn and creating retention plans.
- (Also: pricing strategy and new-product development.)

Q8. What are the limitations of the K-Means algorithm?

- The number of clusters K must be specified in advance.
- Results depend on the random initial centroids (can end in a local optimum); k-means++ and multiple runs help.
- Sensitive to outliers and noise, since means are pulled by extreme values.
- Assumes clusters are roughly spherical and of similar size and density; fails on irregular shapes.
- Works only with numerical data (Euclidean distance) and needs feature scaling.
- Performance degrades in very high-dimensional data.

Q9. How can businesses use customer segmentation to improve marketing and customer retention?

- Send personalized offers and messages instead of one campaign for all, which increases conversion and lowers marketing cost.
- Identify high-value customers and reward them with loyalty programmes.
- Detect low-activity or at-risk segments early and run win-back or retention campaigns.
- Choose the right channel, timing and price sensitivity for each group.
- Improve product development and inventory by understanding each segment's needs, and increase customer lifetime value.

Q10. Compare K-Means Clustering with Hierarchical Clustering based on their working principles and applications.

Aspect	K-Means	Hierarchical clustering
Working principle	Partitional: assigns points to K centroids and iteratively updates them	Builds a tree (dendrogram) by merging closest clusters (agglomerative) or splitting (divisive)
Number of clusters	Must be specified beforehand	Not needed in advance; choose by cutting the dendrogram
Scalability	Fast and efficient for large datasets	Slow, memory-heavy (about $O(n^2)$); suits small to medium data
Result	Can vary with initialization	Deterministic for the same data and linkage
Cluster shape	Spherical clusters	Can handle different shapes depending on linkage
Applications	Customer segmentation, image compression, large datasets	Gene/biological taxonomy, document hierarchy, small datasets where hierarchy matters

Result: K-Means clustering was applied to segment customers into five groups, and marketing strategies were suggested for each segment.